WORKED SOLUTION

STEP III 2011, Question 12

Probability and Statistics • Compound Distributions via PGFs

STEP III 2011, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q12

The key idea

Everything we need about a distribution's mean and variance sits in the first two derivatives of its pgf at $t = 1$. So we differentiate the composite $G(H(t))$ with the chain rule (twice), put $t = 1$, and use $H(1) = 1$. For the coin, N is geometric and each X_i is a single fair-coin indicator, so both pgfs are simple and $G(H(t))$ is a rational function we can expand as a geometric series.

Solution

The general results

For any random variable Z taking non-negative integer values with pgf $P(t) = \mathrm{E}(t^Z)$, we have $P(1) = 1$,

$$P'(1) = \mathrm{E}(Z), \qquad P''(1) = \mathrm{E}(Z(Z-1)),$$

and therefore

$$\mathrm{Var}(Z) = P''(1) + P'(1) - (P'(1))^2.$$

Write $\mu = \mathrm{E}(X_i) = H'(1)$, $\sigma^2 = \mathrm{Var}(X_i)$, and $K(t) = G(H(t))$ for the pgf of Y.

By the chain rule, $K'(t) = G'(H(t))\,H'(t)$. Since $H(1) = 1$,

$$\mathrm{E}(Y) = K'(1) = G'(1)H'(1) = \mathrm{E}(N)\,\mathrm{E}(X_i),$$

which is the first result.

Differentiating again (product rule, then chain rule),

$$K''(t) = G''(H(t))(H'(t))^2 + G'(H(t))\,H''(t),$$

so, again using $H(1) = 1$,

$$K''(1) = G''(1)\mu^2 + \mathrm{E}(N)\,H''(1).$$

From the variance formula above applied to N and to X_i,

$$G''(1) = \mathrm{Var}(N) - \mathrm{E}(N) + (\mathrm{E}(N))^2, \qquad H''(1) = \sigma^2 - \mu + \mu^2.$$

Ask yourself Why not just compute $\mathrm{Var}(Y)$ from $K''(1)$ directly, without rewriting $G''(1)$ and $H''(1)$?

Because the answer we want is in terms of variances, not factorial moments; expressing $G''(1)$ and $H''(1)$ through $\mathrm{Var}(N)$ and σ^2 lets the unwanted terms cancel. Writing $m = \mathrm{E}(N)$,

$$\begin{aligned}\mathrm{Var}(Y) &= K''(1) + K'(1) - (K'(1))^2 \\ &= (\mathrm{Var}(N) - m + m^2)\mu^2 + m(\sigma^2 - \mu + \mu^2) + m\mu - m^2\mu^2 \\ &= \mathrm{Var}(N)\mu^2 + m\sigma^2 + (-m\mu^2 + m\mu^2) + (m^2\mu^2 - m^2\mu^2) + (-m\mu + m\mu) \\ &= \mathrm{Var}(N)(\mathrm{E}(X_i))^2 + \mathrm{E}(N)\,\mathrm{Var}(X_i),\end{aligned}$$

as required.

The coin

The pgf of N. The first head comes on toss n when the first $n-1$ tosses are tails and the nth a head, so $\mathrm{P}(N = n) = (\frac{1}{2})^n$ for $n = 1, 2, 3, \ldots$ (a geometric distribution). Hence, for $|t| < 2$,

$$G(t) = \sum_{n=1}^{\infty} \left(\frac{t}{2}\right)^n = \frac{t/2}{1 - t/2} = \frac{t}{2-t}.$$

The pgf of each X_i. Let $X_i = 1$ if the ith of the further tosses is a head and $X_i = 0$ otherwise. Then $Y = X_1 + \cdots + X_N$, and $H(t) = \frac{1}{2} + \frac{1}{2}t = \frac{1}{2}(1+t)$. The further tosses are independent of each other and of the tosses that decided N, so the conditions of the general result hold.

The pgf of Y.

$$G(H(t)) = \frac{\frac{1}{2}(1+t)}{2 - \frac{1}{2}(1+t)} = \frac{1+t}{3-t}.$$

Mean and variance. For the geometric distribution with $p = \frac{1}{2}$, $\mathrm{E}(N) = 1/p = 2$ and $\mathrm{Var}(N) = (1-p)/p^2 = 2$ (or differentiate G: $G'(t) = 2/(2-t)^2$, $G''(t) = 4/(2-t)^3$, giving $G'(1) = 2$, $G''(1) = 4$ and $\mathrm{Var}(N) = 4 + 2 - 4 = 2$). Each X_i is Bernoulli$(\frac{1}{2})$, so $\mathrm{E}(X_i) = \frac{1}{2}$ and $\mathrm{Var}(X_i) = \frac{1}{4}$. Therefore

$$\mathrm{E}(Y) = 2 \cdot \tfrac{1}{2} = 1, \qquad \mathrm{Var}(Y) = 2 \cdot \tfrac{1}{4} + 2 \cdot \tfrac{1}{4} = 1.$$

The distribution of Y. For $|t| < 3$ we expand

$$\frac{1+t}{3-t} = \frac{1+t}{3} \cdot \frac{1}{1 - t/3} = \frac{1+t}{3} \sum_{k=0}^{\infty} \frac{t^k}{3^k} = \sum_{k=0}^{\infty} \frac{t^k}{3^{k+1}} + \sum_{k=0}^{\infty} \frac{t^{k+1}}{3^{k+1}}.$$

The coefficient of t^0 comes only from the first sum. For $r \geq 1$ the coefficient of t^r is $\dfrac{1}{3^{r+1}} + \dfrac{1}{3^r} = \dfrac{4}{3^{r+1}}$. So

$$\mathrm{P}(Y = 0) = \frac{1}{3}, \qquad \mathrm{P}(Y = r) = \frac{4}{3^{r+1}} \quad (r = 1, 2, 3, \ldots).$$

Checks

- The probabilities sum to 1: $\frac{1}{3} + 4\sum_{r\geq 1} 3^{-(r+1)} = \frac{1}{3} + 4 \cdot \frac{1}{6} = 1$.
- $\mathrm{P}(Y = 0)$ directly: $\sum_{n\geq 1} \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n = \sum_{n\geq 1} 4^{-n} = \frac{1}{3}$. Similarly $\mathrm{P}(Y = 1) = \sum_{n\geq 1} n\, 4^{-n} = \frac{4}{9}$, agreeing with $4/3^2$.
- Summing $\mathrm{P}(N = n)\binom{n}{r}2^{-n}$ exactly over $n \leq 200$ reproduced $\mathrm{P}(Y = r)$ for $r \leq 7$, and gave mean and variance both 1.000; a simulation of 200 000 runs gave 0.996 and 0.994.

Common slips

- Taking N to start at 0 (pgf $1/(2 - t)$): here N counts the toss that gives the head, so $N \geq 1$.
- Forgetting that $P''(1)$ is $\mathrm{E}(Z^2) - \mathrm{E}(Z)$, not $\mathrm{E}(Z^2)$, and so losing the $+P'(1)$ in the variance formula.
- Giving $4/3^{r+1}$ for all $r \geq 0$: the $r = 0$ term is different.

Beyond STEP

Y is a *random sum* (a compound distribution), and the variance formula is a special case of the law of total variance, $\mathrm{Var}(Y) = \mathrm{E}\big(\mathrm{Var}(Y \mid N)\big) + \mathrm{Var}\big(\mathrm{E}(Y \mid N)\big)$. Compositions of pgfs are also the engine of branching processes, where $G(G(\cdots G(t)))$ gives the size of the nth generation and the chance of extinction is a fixed point of G. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions, random sums and branching processes at exactly this level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.